

Question ID: 78d5f91a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$f(x) = x^3 + 3x^2 - 6x - 1$

For the function f defined above, what is the value of $f(-1)$?

Answer

- A. -11
- B. -7
- C. 7
- D. 11

Correct Answer: C

Rationale

Choice C is correct. Substituting -1 for x in the given function f gives $f(-1) = (-1)^3 + 3(-1)^2 - 6(-1) - 1$, which simplifies to $f(-1) = -1 + 3(1) - 6(-1) - 1$. This further simplifies to $f(-1) = -1 + 3 + 6 - 1$, or $f(-1) = 7$.

Choice A is incorrect and may result from correctly substituting -1 for x in the function but incorrectly simplifying the resulting expression to $f(-1) = -1 - 3 - 6 - 1$, or -11 . Choice B is incorrect and may result from arithmetic errors. Choice D is incorrect and may result from correctly substituting -1 for x in the function but incorrectly simplifying the expression to $f(-1) = 1 + 3 + 6 + 1$, or 11.